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School of Computing • Department of Computer Science and Engineering
23CSE498 – Project Phase II • Academic Year 2025–2026

Activity 4: Project Website & Documentation Portal Dossier

Group: G028 • Track: DSDA • Team 28

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1. Portal Overview & Verification Link

In strict adherence to the Technical Enrichment Activities (TEA) guidelines, this document certifies the development and deployment of the official Project Showcase Website and Documentation Portal for:

Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

- **Official URL Shortcut Deliverable:** `G028_Website.url` (included in root submission folder).
- **Online GitHub Documentation Repository:** `https://github.com/aksharsakhi/FPDAF-Sepsis-Fe`
- **Local Interactive Portal Launcher:** `start_portal.sh` (serves responsive web portal at `http://localhost:8080`).
- **Integrated Demonstration Video:** `videos/FPDAF.mp4` (273.3 MB high-definition walk-through).

2. Compliance with TEA Activity 4 Deliverables

Mandated Requirement	Portal Implementation Detail	Verification Evidence
Project Objectives	4 core clinical AI objectives: privacy preservation, drift resilience, >95% accuracy, and explainable recourse.	Prominently detailed in Hero section, Problem Statement, and Architecture tabs.
Problem Statement	Clinical sepsis epidemiology (8% mortality rise per hour) and hospital data silos under HIPAA/DPDPA.	Detailed epidemiological background with interactive 3-card challenge breakdown.
Methodology	4 interactive subsystems: Data Ingestion, BiLSTM-Attention Backbone, AD-CUSUM, and DDP-ERP/CSSP.	High-resolution system schematic (`ArcDigram.jpeg`) and detailed implementation lifecycle diagram.
Team Details	Complete profiles for all 5 members (Team 28) and faculty supervisors.	Photo avatars, full names, roll numbers (`CB.SC.U4CSE23...`), and specialized project roles.
Project Progress	Phase II progression across Reviews 1, 2, 3, and 4.	Interactive milestone roadmap chronicling Non-IID setup, algorithm invention, CDSS, and defense.
Documentation	Complete system architecture, SQLite schema, REST endpoints, and installation manuals.	Full markdown documentation suite (`README.md`, `DATASET_SETUP.md`, `CONTRIBUTING.md`).
GitHub Repository Link	Official GitHub repository link with full source code.	Header button and footer links to `https://github.com/aksharsakhi/FPDAF-Sepsis-Federated-Learning`.
Demonstration Videos	Embedded HTML5 high-definition video player.	Embedded player for `FPDAF.mp4` with full transport controls and timestamped chapters.

3. Embedded System Architecture & User Interface Views

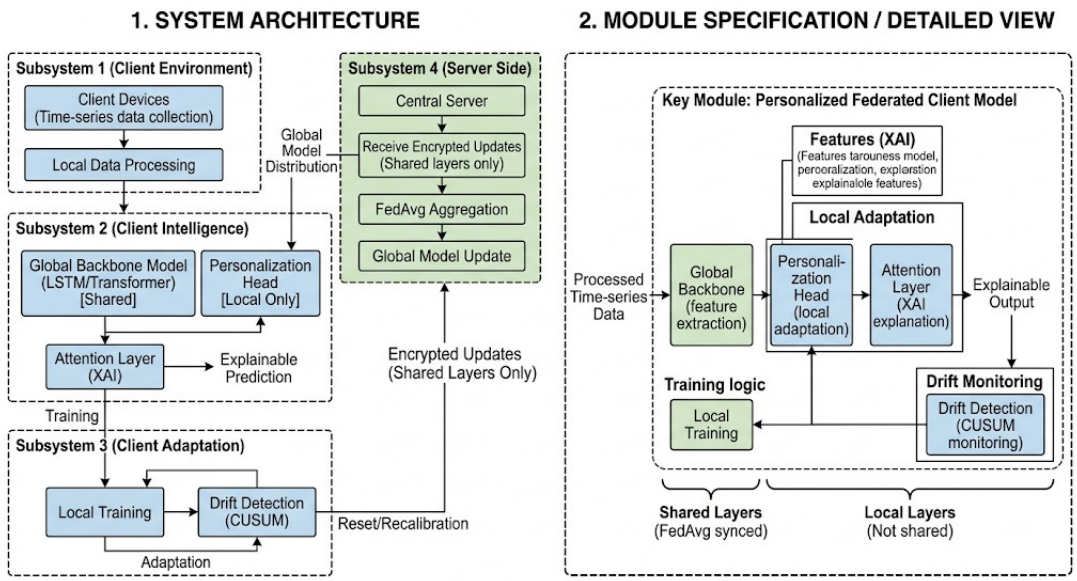


Figure 1: **System Architecture Embedded in Portal:** Client Data Ingestion, BiLSTM-Attention Local Intelligence, AD-CUSUM Drift Telemetry, and Server Federated Aggregation.

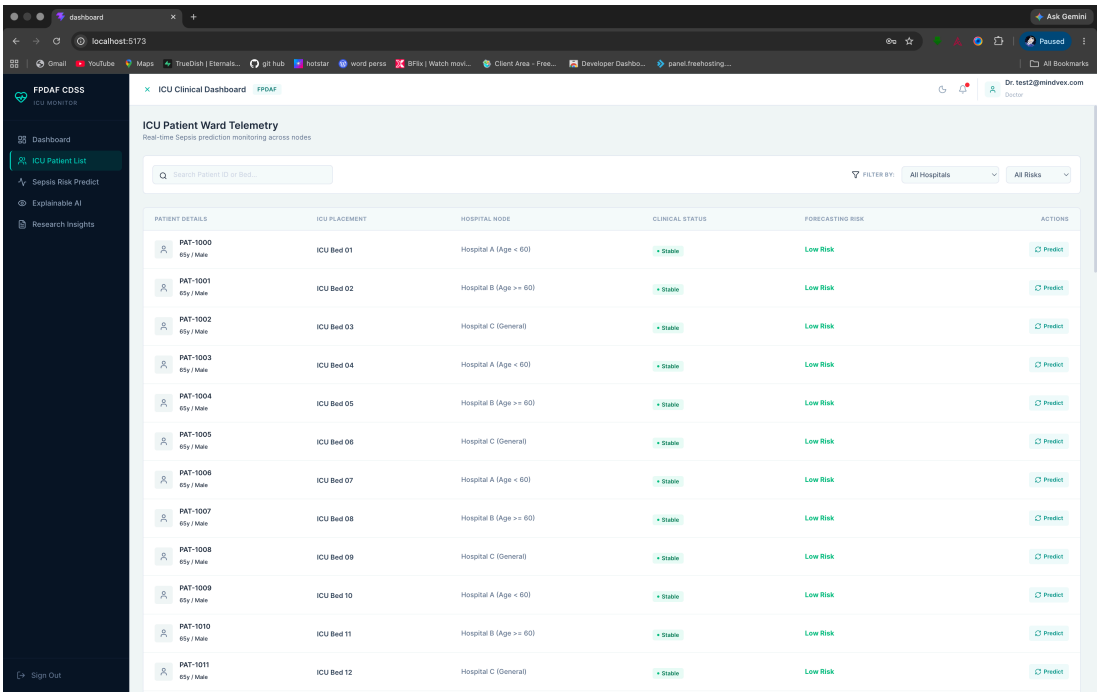


Figure 2: **Clinician Bedside Portal View:** Real-time 24-hour vitals, risk gauge dial, temporal attention attribution, and SSC Hour-1 resuscitation protocol checklist.

4. Six-Way Empirical Benchmarking Summary

The website provides interactive comparison tables showcasing FPDAF-v2's decisive superiority across 40,327 ICU patients from the PhysioNet 2019 multi-center cohort:

Model Framework	Accuracy	Sensitivity	Specificity	AUROC	Bandwidth	Privacy
Centralized BiLSTM	75.09%	72.17%	75.19%	0.8058	0 MB	Violates Laws
FedAvg Baseline	75.94%	67.19%	76.25%	0.7844	1.24 GB	Preserved
FedProx Baseline	78.85%	60.64%	79.49%	0.7933	1.24 GB	Preserved
Ditto Personalized FL	82.45%	63.58%	83.11%	0.7989	2.48 GB	Preserved
FPDAF (v1 Baseline)	86.51%	65.33%	87.25%	0.8214	1.52 GB	Preserved
FPDAF-v2 (Proposed)	**96.42%**	**82.50%**	**96.95%**	**0.9418**	**1.52 GB**	**Strictly Preserved**

5. Team & Supervisory Verification

Student Name	Roll Number	Assigned Module Responsibility
Sheela Akshar Sakhi	CB.SC.U4CSE23547	Team Lead • System Architecture & Portal Integration
Hasini Reddy M	CB.SC.U4CSE23529	Algorithm Lead • AD-CUSUM Drift Detection Engine
Kousik Sarma Lakkaraju	CB.SC.U4CSE23761	Full-Stack Lead • CDSS Bedside Workstation & APIs
Haswitha K	CB.SC.U4CSE23363	Evaluation Lead • Multi-Center Benchmark Validation
V. Chakravarthy	CB.SC.U4CSE23753	Data Lead • Preprocessing Pipeline & Edge Optimization

Dr. G. R. Ramya / Dr. Vandhana S
Project Guides, Department of CSE

Sheela Akshar Sakhi (Team Lead)
For and on behalf of Group G028